SentinelAI

Transforming Every Streetlight into a Silent Guardian

AI-Driven Distributed Public Safety Mesh for Smart Cities

CONCEPT-TO-PROTOTYPE EXECUTION PLAN

Sai Nirisha Chowdary Pavuluri • Saatwik Sairaam Vasamsetti
VIT-AP University · July 2026

THE PROBLEM

Help arrives too late — because no one is watching

1.68 L

road-accident deaths in India in 2022 — many die waiting for help that comes too late

MoRTH, Road Accidents in India 2022 (1,68,491)

50%

of road-accident fatalities could be saved with timely medical care in the golden hour

Law Commission of India, Report 201 (2006)

4.4 L+

crimes against women reported annually; isolated public spaces remain unmonitored

NCRB Crime in India 2022

Why current solutions fail

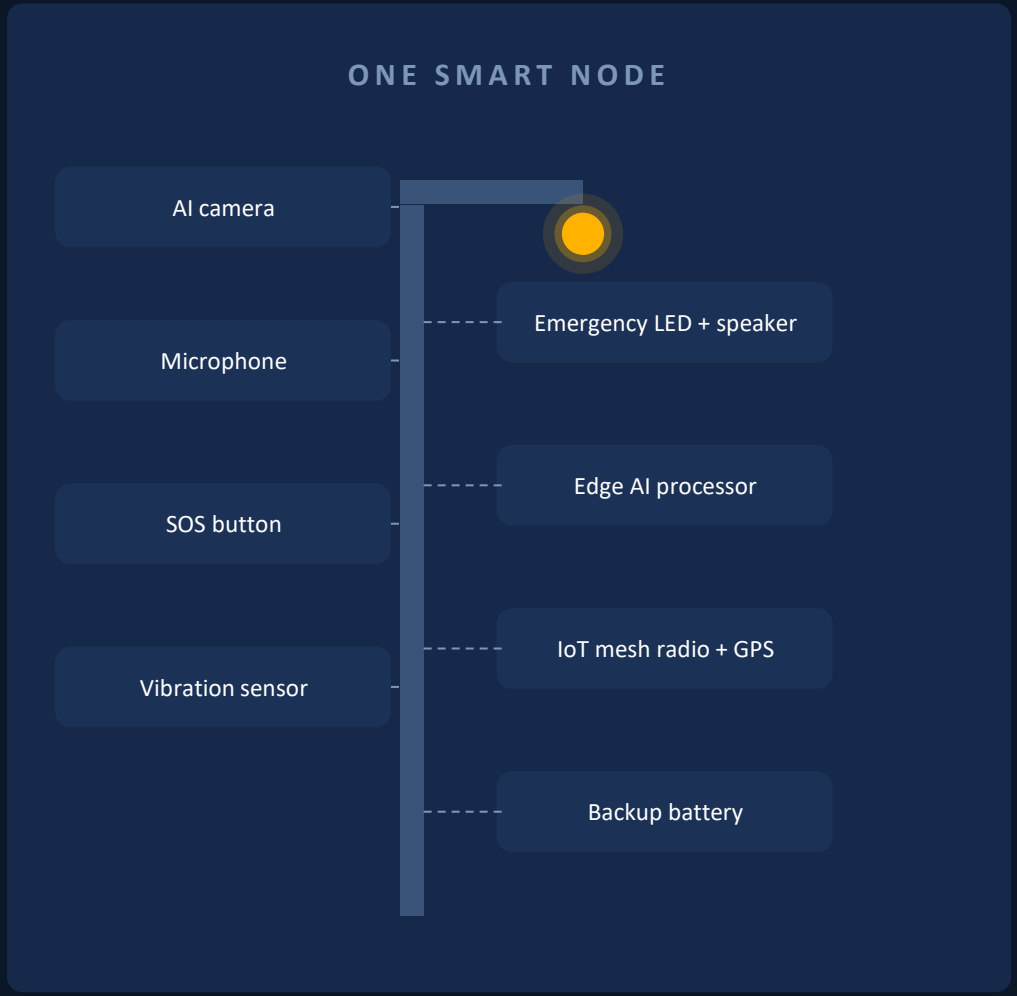
- **CCTV cameras —** only record incidents — nobody responds in real time
- **Public infrastructure —** is passive — streetlights illuminate, they don't protect
- **Emergency apps —** require a conscious victim to manually ask for help
- **Emergency services —** receive late, incomplete, unverified information

THE SOLUTION

Infrastructure that detects, verifies and responds

SentinelAI retrofits existing streetlights, bus stops and public poles into AI-powered emergency response nodes. Each node monitors its surroundings with fused sensors and on-device AI — no victim action required.

1. **DETECT** AI camera, microphone and vibration sensor spot accidents and distress automatically
2. **VERIFY** Multi-sensor fusion confirms a real emergency before anyone is alerted — cutting false alarms
3. **RESPOND** Police, ambulance and family get verified alerts with exact GPS location in seconds
4. **PROTECT** Lights brighten, cameras track, evidence is stored securely, responders get the safest route



CORE CAPABILITIES

Five systems, one guardian

Accident Detection

Detects vehicle, bike and pedestrian collisions by fusing AI vision, crash-sound signatures and road vibration — then auto-dispatches help with GPS location.

Women's Safety

SOS press, a scream, or AI-detected suspicious activity triggers police + family alerts, floodlighting, active camera tracking and secure evidence capture.

Safe Corridor

Neighbouring poles coordinate as a mesh: lights brighten along the victim's path, cameras hand off tracking, responders receive the fastest safe route.

Edge AI

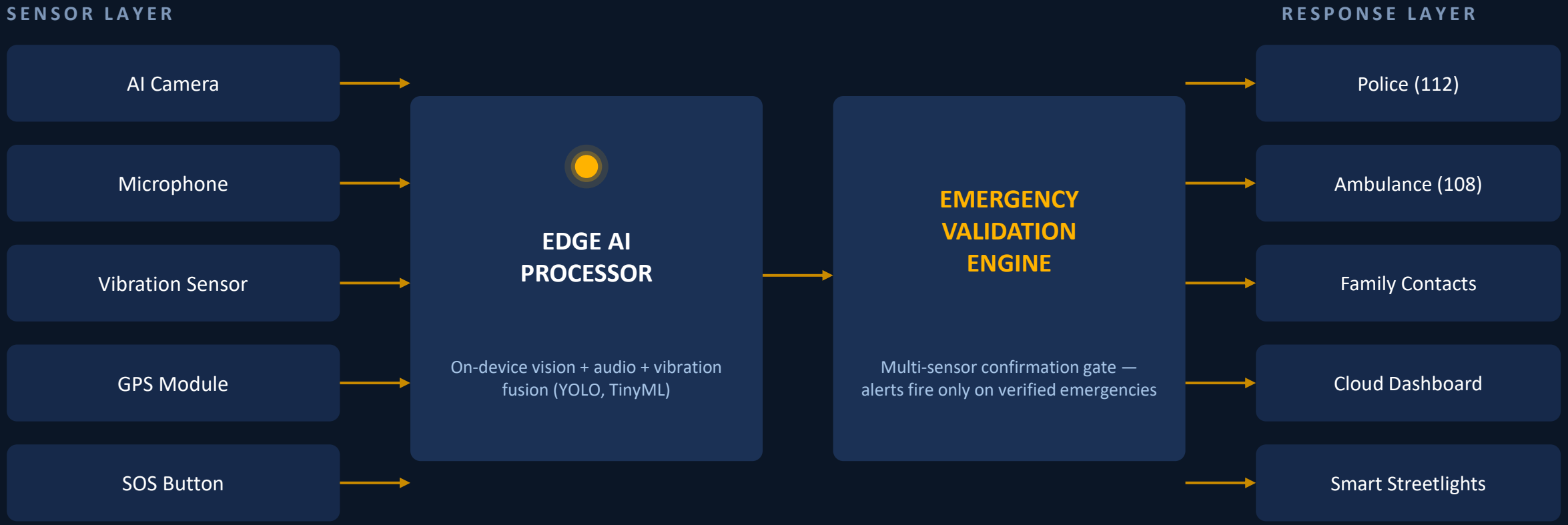
Video and audio are processed on the pole, not in the cloud — sub-second response, minimal bandwidth, and raw footage never leaves the node by default.

Predictive Analytics

Learns from every event to flag accident-prone roads, unsafe zones, repeat crime hotspots and failing infrastructure before incidents happen.

SYSTEM ARCHITECTURE

Sense → verify on the edge → dispatch



Privacy by design: all continuous analysis happens on the pole. Footage is encrypted and leaves the node only when a verified emergency occurs.

TECHNOLOGY

Proven components, novel combination

HARDWARE — PER NODE

- AI camera (night-capable)
- Microphone array
- Vibration / impact sensor
- GPS module
- Physical SOS button
- Emergency LED floodlight
- Public-address speaker
- Edge AI processor (Jetson-class)
- IoT mesh radio (LoRa / NB-IoT)
- Backup battery (8+ hr)

SOFTWARE & AI

AI / Vision YOLO, OpenCV, TensorFlow, Edge Impulse (TinyML)

Edge runtime On-device inference; event-only cloud sync

Backend Node.js + Express REST APIs

Database MongoDB (events, nodes, analytics)

Dashboard React + Tailwind CSS control room

Cloud AWS (IoT Core, S3 evidence vault)

Alerts Firebase push + SMS gateway to 112/108

Mapping Google Maps — live node + incident map

INNOVATION

Records vs. responds

	CCTV	SOS Apps	SentinelAI
Detects incidents automatically	✗	✗	✓
Works when victim is unconscious	✗	✗	✓
Verifies before alerting (low false alarms)	✗	✗	✓
Coordinates across locations (mesh)	✗	✗	✓
Active deterrence (light + speaker)	✗	✗	✓
Evidence capture	✓	✗	✓

CANDIDATE IP AREAS

- Multi-sensor emergency validation algorithm
- Distributed pole-to-pole AI coordination
- AI-driven Safe Corridor activation
- Edge-based emergency detection framework

Candidate areas only — no prior-art search has been run yet; that search is a Phase 1 deliverable. The SentinelAI column describes the designed system, not a built one.

EXECUTION PLAN

Concept to prototype in 8 months — then scale

1 2 3 4 5

RESEARCH & VALIDATION

Months 1–2

- Interviews: police, 108 operators, municipal officials, 50+ commuters
- Freeze requirements + success metrics
- Prior-art search on 4 IP areas

PROTOTYPE BUILD

Months 3–5

- Single working node: sensor fusion + edge AI on Jetson
- Crash / scream / SOS detection models trained
- Dashboard MVP + SMS-alert loop

CAMPUS PILOT

Months 6–8

- 5-node mesh on campus roads
- Safe Corridor live demo
- False-alarm rate tuned < 5%

CITY PILOT

Months 9–14

- 25–50 nodes with municipal partner
- Police control-room integration
- Reliability + weather hardening

COMMERCIAL

Month 15+

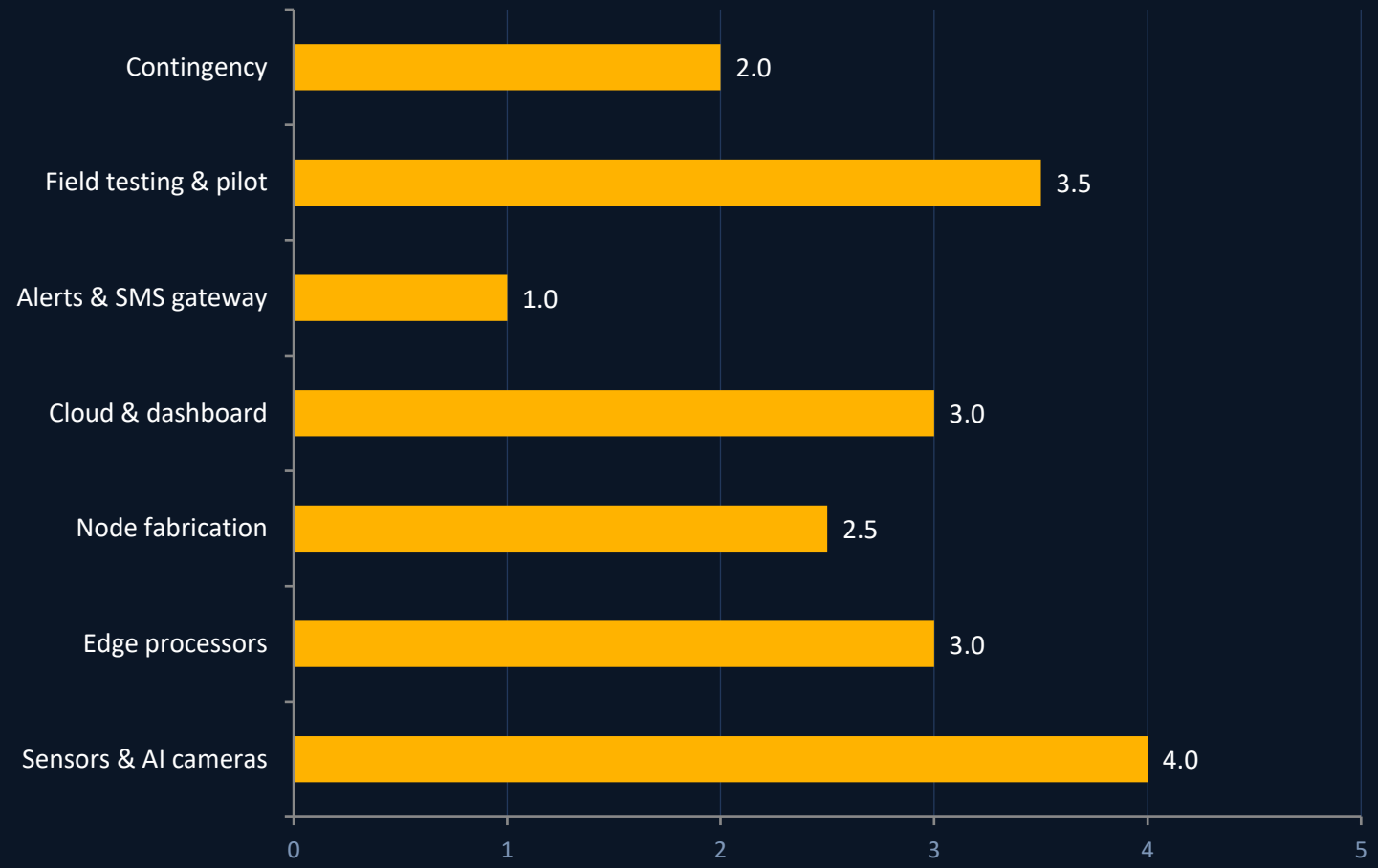
- Smart City tenders + NHAI corridors
- SaaS dashboard subscriptions
- Manufacturing partnerships

THIS SUBMISSION FUNDS PHASES 1–3: WORKING MESH PROTOTYPE IN 8 MONTHS

Gate to next phase: each phase ends with a demo + metric review

PROTOTYPE BUDGET

₹19 lakh to a working 5-node mesh



TOTAL PROTOTYPE ESTIMATE

₹18–20 L

Phases 1–3 · 8 months · 5-node campus mesh

COST DISCIPLINE

- Commodity hardware only — no custom silicon
- Retrofit existing poles: zero civil-works cost
- Open-source AI stack (YOLO, OpenCV, TF)
- Campus pilot before any city spend

Hardware lines (₹9.5 L) cover ~11 node-builds at the ₹82 k per-node BOM on the next slide — 5 deployed, plus a lab reference unit, two build iterations and spare cameras/processors.

UNIT ECONOMICS — INDICATIVE

What one node costs, and what it earns

PER-NODE BILL OF MATERIALS (PROTOTYPE)

AI camera (night-capable)	₹ 8,000
Edge AI processor (Jetson-class)	₹ 35,000
Mic array + vibration + GPS sensors	₹ 6,000
LoRa / NB-IoT radio module	₹ 3,000
Backup battery + power driver	₹ 7,000
IP66 enclosure, LED floodlight, speaker, mount	₹ 14,000
Assembly, wiring & bench test	₹ 9,000
TOTAL PARTS COST / NODE	**≈ ₹ 82,000**

AT VOLUME (1,000+ NODES)

₹ 48–55 k

Integrated carrier board replaces dev-kit; 35–40% volume pricing on sensors and enclosure

INDICATIVE PRICING PER NODE

- **Hardware + installation** ₹ 95 k – 1.2 L one-time
- **Maintenance (AMC)** ₹ 8–10 k / node / year
- **SaaS dashboard + analytics** per city / campus licence

Component prices are indicative (public market pricing, July 2026) — replaced with vendor quotes in Phase 1. Hardware margins at volume fund the recurring SaaS + AMC engine.

MARKET & BUSINESS MODEL

Built for India's municipal safety infrastructure

WHO BUYS

- **Smart City SPVs & municipal corporations**
  100 cities were selected under the Smart Cities Mission
- **NHAI & state highway authorities**
  black-spot corridors, toll plazas
- **Metro rail, airports & universities**
  campuses with existing pole infrastructure
- **Private townships & industrial parks**
  safety as a premium amenity

Beneficiaries: women, senior citizens, students, daily commuters, highway travellers

HOW WE EARN

Hardware + installation	one-time, per smart node
SaaS dashboard	annual subscription per city / campus
Maintenance (AMC)	recurring, per node per year
AI analytics reports	monthly hotspot & infrastructure insights
API licensing	incident feed for maps, insurers, fleets

Future: Middle East, Southeast Asia, Europe

DEMONSTRATION EVIDENCE

Interactive simulation of the mesh in action

We built an interactive software simulation of a SentinelAI deployment — a live 3D city block of smart nodes that walks through the full incident flow: a scripted crash and SOS event, the multi-sensor validation gate, Safe Corridor activation and dispatch, end to end.

Live node mesh

9 smart poles on a 3D city block, each streaming simulated sensor state

Incident walkthrough

crash + distress events step through the validation gate and IR camera view

Safe Corridor

watch neighbouring poles brighten and hand off camera tracking

Predictive layer

30-day risk-forecast heatmap built from mesh history

Scripted simulation — synthetic sensor values, no trained model or hardware yet. It demonstrates the intended flow and interface, not detection accuracy.

sentinelai/demo/index.html (live, any browser) · SentinelAI_Demo.mp4 (recorded)

DEMO VIDEO

[paste demo video link here]

≈1-minute screen recording of the auto demo: crash flow, women's-safety SOS flow, then the 30-day predictive-risk heatmap.

Hardware node demo planned at end of Phase 2.

TEAM

Builders with shipped projects

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Sai Nirisha Chowdary Pavuluri

Founder

- Integrated M.Tech, CSE (5-year) — VIT-AP University, Batch 2024–29
- Concept originator — authored the SentinelAI vision, market analysis and IP strategy; built a GPT implementation from scratch in pure Python
- Owns: product vision, stakeholder validation, business model, partnerships

GitHub /Nirisha22

LinkedIn /nirisha-chowdary

S

Saatwik Sairaam Vasamsetti

Co-Founder & Tech Lead

- Integrated M.Tech, CSE (5-year) — VIT-AP University, Batch 2024–29
- Full-stack developer — shipped PathAhead (career-guidance platform), an ML stock-prediction system, and multiple production web platforms
- Owns: technical architecture, edge AI models, sensor fusion, dashboard build

GitHub /saatwik-1157

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ADVISORS & VALIDATION NETWORK (IN PROGRESS)

Phase 1 targets: one police-department mentor, one 108/EMRI operations contact, one municipal smart-city officer, one faculty advisor (embedded systems / AI) at VIT-AP. Names added as they are confirmed.

RISKS & MITIGATION

We designed for the failure modes

Privacy concerns

Edge-first processing — footage stays on the node; encrypted, event-only uploads; DPDP-Act-aligned data policy; no facial recognition in v1

False alarms

Multi-sensor validation gate (vision + audio + vibration must agree); campus pilot tunes threshold to < 5% before any city deployment

Weather & durability

IP66-rated enclosures; sensor redundancy; degraded-mode operation when any one sensor fails

Hardware maintenance

Self-diagnostics heartbeat to dashboard; AMC model funds field service; modular swap-in parts

Government approvals

Pilot under university MoU first; align with Smart Cities Mission procurement; police partnership from Phase 1

Installation cost

Retrofit existing poles — no new civil works; commodity hardware; per-node cost drops with volume

Every pole we light up is a life we might save.

SentinelAI makes public infrastructure aware, responsive and connected — faster emergency response, safer streets, smarter cities.

SDG 3
Good Health

SDG 5
Gender Equality

SDG 9
Industry & Innovation

SDG 11
Sustainable Cities

SDG 16
Peace & Justice

THE ASK

₹18–20 lakh + incubation support to deliver a working 5-node campus mesh in 8 months — validated with police, ambulance operators and municipal officials.

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